FIELDS GENERATED BY P-POWER ROOTS WITH BOUNDED DENOMINATORS AND PURELY INSEPARABLE EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Purely inseparable elements as p-power roots. Used inside the counted proof at lines 11080-11081. Identifies an element whose p^n-th power lies in k as purely inseparable over k in characteristic p.
- The p-power map in characteristic different from p. Used at proof line 11079. Shows that when the characteristic is not p the generator condition collapses to membership in k.

Lemma 0.1. *Let k'/k be a field extension. Let p be a prime number. The following are equivalent*

(1) *k' is generated as a field extension of k by elements x such that there exists an $n > 0$ with $x^{p^n} \in k$ and $p^n x \in k$, and*
(2) *$k = k'$ or the characteristic of k and k' is p and k'/k is purely inseparable.*

Proof. Let $x \in k'$. If there exists an $n > 0$ with $x^{p^n} \in k$ and $p^n x \in k$ and if the characteristic is not p, then $x \in k$. If the characteristic is p, then we find $x^{p^n} \in k$ and hence x is purely inseparable over k. □

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